Task 4: Loan Default Risk with Business Cost Optimization

Problem Statement and Objective

Objective

The goal of this project is to predict whether an applicant will default on a loan. Beyond simple classification, we aim to minimize the total business cost. We will achieve this by:

- Cleaning and preprocessing the Home Credit dataset.
- Training a classification model.
- Defining a cost function where a **False Negative** (not predicting a default that happens) is significantly more expensive than a **False Positive** (wrongly flagging a good client).
- Optimizing the decision threshold to find the **"Sweet Spot"** for the bank's profitability.

```
# This Python 3 environment comes with many helpful analytics
libraries installed
# It is defined by the kaggle/python Docker image:
https://github.com/kaggle/docker-python
# For example, here's several helpful packages to load

import numpy as np # linear algebra
import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv)

# Input data files are available in the read-only "../input/"
directory
# For example, running this (by clicking run or pressing Shift+Enter)
will list all files under the input directory

import os
for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))

# You can write up to 20GB to the current directory (/kaggle/working/)
that gets preserved as output when you create a version using "Save &
Run All"
# You can also write temporary files to /kaggle/temp/, but they won't
be saved outside of the current session
```

Dataset Loading and Initial Setup

We'll start by importing the necessary libraries and loading the main application file.

```
import pandas as pd
import numpy as np
import os
# We will use os.walk to find the exact location of
application_train.csv automatically
target_file = 'application_train.csv'
file_path = None

for root, dirs, files in os.walk(path):
    if target_file in files:
        file_path = os.path.join(root, target_file)
        break

if file_path:
    df = pd.read_csv(file_path)
    print(f"Success! Dataset loaded from: {file_path}")
    print(f"Dataset Shape: {df.shape}")
else:
    print(f"Error: Could not find {target_file} in {path}")
    # Printing directory content to help you debug
    print("Files found in path:", os.listdir(path))

df.head()
```

```
Success! Dataset loaded from:
/kaggle/input/datasets/megancrenshaw/home-credit-default-risk/home-
credit-default-risk/application_train.csv
Dataset Shape: (307511, 122)

   SK_ID_CURR  TARGET NAME_CONTRACT_TYPE CODE_GENDER FLAG_OWN_CAR  \
0      100002       1         Cash loans           M            N
1      100003       0         Cash loans           F            N
2      100004       0    Revolving loans           M            Y
3      100006       0         Cash loans           F            N
4      100007       0         Cash loans           M            N

  FLAG_OWN_REALTY  CNT_CHILDREN  AMT_INCOME_TOTAL  AMT_CREDIT
AMT_ANNUITY  \
0               Y             0          202500.0    406597.5
24700.5
1               N             0          270000.0   1293502.5
35698.5
2               Y             0           67500.0    135000.0
6750.0
3               Y             0          135000.0    312682.5
29686.5
```

```
4                Y             0          121500.0    513000.0
21865.5

   ...  FLAG_DOCUMENT_18 FLAG_DOCUMENT_19 FLAG_DOCUMENT_20
FLAG_DOCUMENT_21  \
0  ...                 0                0                0
0
1  ...                 0                0                0
0
2  ...                 0                0                0
0
3  ...                 0                0                0
0
4  ...                 0                0                0
0

  AMT_REQ_CREDIT_BUREAU_HOUR AMT_REQ_CREDIT_BUREAU_DAY  \
0                        0.0                       0.0
1                        0.0                       0.0
2                        0.0                       0.0
3                        NaN                       NaN
4                        0.0                       0.0

   AMT_REQ_CREDIT_BUREAU_WEEK  AMT_REQ_CREDIT_BUREAU_MON  \
0                         0.0                        0.0
1                         0.0                        0.0
2                         0.0                        0.0
3                         NaN                        NaN
4                         0.0                        0.0

   AMT_REQ_CREDIT_BUREAU_QRT  AMT_REQ_CREDIT_BUREAU_YEAR
0                        0.0                        1.0
1                        0.0                        0.0
2                        0.0                        0.0
3                        NaN                        NaN
4                        0.0                        0.0

[5 rows x 122 columns]
```

Data Cleaning and Preprocessing

Financial datasets are notorious for missing values and categorical features. We will handle these to make the data "model-ready."

```
# 1. Handling Missing Values: Imputing median for numerical and
'Unknown' for categorical
num_cols = df.select_dtypes(include=['float64', 'int64']).columns
cat_cols = df.select_dtypes(include=['object']).columns
```

```
# Drop columns with more than 50% missing values to reduce noise
df = df.dropna(thresh=df.shape[0]*0.5, axis=1)

# Impute numericals with median
imputer_num = SimpleImputer(strategy='median')
df[df.select_dtypes(include=[np.number]).columns] =
imputer_num.fit_transform(df.select_dtypes(include=[np.number]))

# 2. Encoding Categorical Variables
# Label encoding for binary, One-Hot for others
le = LabelEncoder()
for col in cat_cols:
    if col in df.columns:
        if len(df[col].unique()) <= 2:
            df[col] = le.fit_transform(df[col].astype(str))

# One-hot encoding for remaining categorical features
df = pd.get_dummies(df)

print("Preprocessing Complete. New Shape:", df.shape)
```

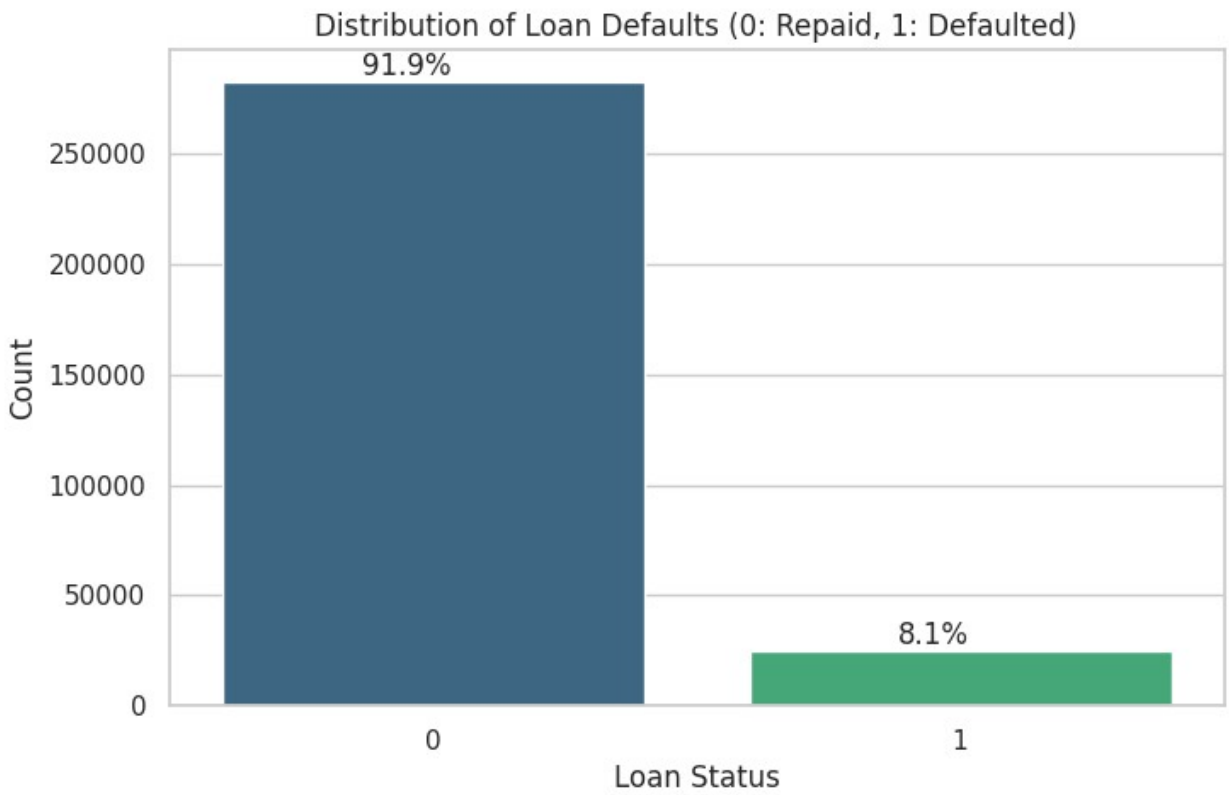
```
Preprocessing Complete. New Shape: (307511, 191)
```

Exploratory Data Analysis (EDA)

Understanding the target distribution is key. Loan default datasets are usually highly imbalanced.

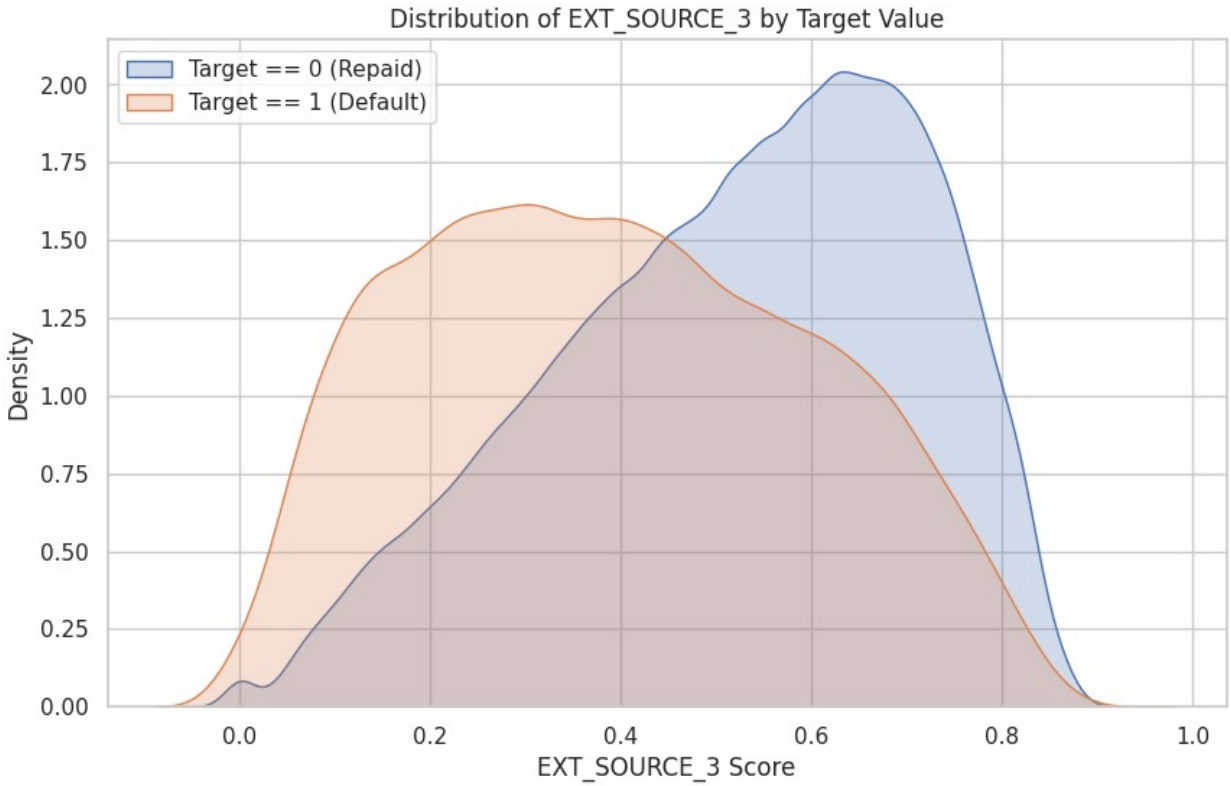
```
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

# Ensure the plotting style is set
sns.set_theme(style="whitegrid")

# 1. Visualize Target Distribution
plt.figure(figsize=(8, 5))
# Fixed the warning by assigning x to hue and setting legend=False
ax = sns.countplot(x='TARGET', data=df, hue='TARGET',
palette='viridis', legend=False)
plt.title('Distribution of Loan Defaults (0: Repaid, 1: Defaulted)')
plt.xlabel('Loan Status')
plt.ylabel('Count')

# Add percentage labels on top of bars
total = len(df['TARGET'])
for p in ax.patches:
    height = p.get_height()
    if height > 0:
```

```
        percentage = '{:.1f}%'.format(100 * height/total)
        x = p.get_x() + p.get_width() / 2 - 0.05
        y = height
        ax.annotate(percentage, (x, y), size = 12, ha='center',
va='bottom')

plt.show()

# 2. Correlation Analysis
print("Calculating correlations with TARGET...")
# Fix: Added numeric_only=True to ignore string columns like 'Cash
loans'
correlations = df.corr(numeric_only=True)['TARGET'].sort_values()

print("\n--- Top 10 Features Most Correlated with Default (Target 1)
---")
print(correlations.tail(11)[:-1]) # Excluding TARGET itself

print("\n--- Top 10 Features Most Correlated with Repayment (Target 0)
---")
print(correlations.head(10))

# 3. Visualizing a Key Feature (e.g., EXT_SOURCE_3)
if 'EXT_SOURCE_3' in df.columns:
    plt.figure(figsize=(10, 6))
    sns.kdeplot(df.loc[df['TARGET'] == 0, 'EXT_SOURCE_3'],
label='Target == 0 (Repaid)', fill=True)
    sns.kdeplot(df.loc[df['TARGET'] == 1, 'EXT_SOURCE_3'],
label='Target == 1 (Default)', fill=True)
    plt.title('Distribution of EXT_SOURCE_3 by Target Value')
    plt.xlabel('EXT_SOURCE_3 Score')
    plt.ylabel('Density')
    plt.legend()
    plt.show()
else:
    print("\nNote: EXT_SOURCE_3 column not found for visualization.")
```




```
Calculating correlations with TARGET...

--- Top 10 Features Most Correlated with Default (Target 1) ---
DAYS_REGISTRATION              0.041975
FLAG_DOCUMENT_3                0.044346
REG_CITY_NOT_LIVE_CITY         0.044395
FLAG_EMP_PHONE                 0.045982
REG_CITY_NOT_WORK_CITY         0.050994
DAYS_ID_PUBLISH                0.051457
DAYS_LAST_PHONE_CHANGE         0.055218
REGION_RATING_CLIENT           0.058899
REGION_RATING_CLIENT_W_CITY    0.060893
DAYS_BIRTH                     0.078239
Name: TARGET, dtype: float64

--- Top 10 Features Most Correlated with Repayment (Target 0) ---
EXT_SOURCE_3                 -0.178919
EXT_SOURCE_2                 -0.160472
EXT_SOURCE_1                 -0.155317
DAYS_EMPLOYED                -0.044932
FLOORSMAX_AVG                -0.044003
FLOORSMAX_MEDI               -0.043768
FLOORSMAX_MODE               -0.043226
AMT_GOODS_PRICE              -0.039645
REGION_POPULATION_RELATIVE   -0.037227
```

```
ELEVATORS_AVG                 -0.034199
Name: TARGET, dtype: float64
```



Model Building (Logistic Regression)

We will use Logistic Regression as our baseline. It provides probability scores which are essential for threshold optimization.

```
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import roc_auc_score
from sklearn.impute import SimpleImputer

# 1. Final Feature Selection
# Drop target and ID, and keep only numeric columns
X = df.drop(columns=['TARGET', 'SK_ID_CURR'],
errors='ignore').select_dtypes(include=[np.number])
y = df['TARGET']

# 2. THE FIX: Global Imputer
# This ensures that even if get_dummies or earlier steps created NaNs,
they are filled now.
final_imputer = SimpleImputer(strategy='median')
```

```
X_clean = final_imputer.fit_transform(X)

print(f"Features used for training: {X_clean.shape[1]}")

# 3. Split the data
X_train, X_test, y_train, y_test = train_test_split(
    X_clean, y, test_size=0.2, random_state=42, stratify=y
)

# 4. Scaling
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# 5. Model Training
# 'balanced' weights help with the 90/10 split of non-defaulters vs
defaulters
model = LogisticRegression(max_iter=1000, class_weight='balanced',
random_state=42)
model.fit(X_train_scaled, y_train)

# 6. Get Probabilities
y_probs = model.predict_proba(X_test_scaled)[:, 1]

print("Model training successful! No NaNs detected.")
print(f"ROC AUC Score: {roc_auc_score(y_test, y_probs):.4f}")
```

```
Features used for training: 104
Model training successful! No NaNs detected.
ROC AUC Score: 0.7377
```

Business Cost Optimization

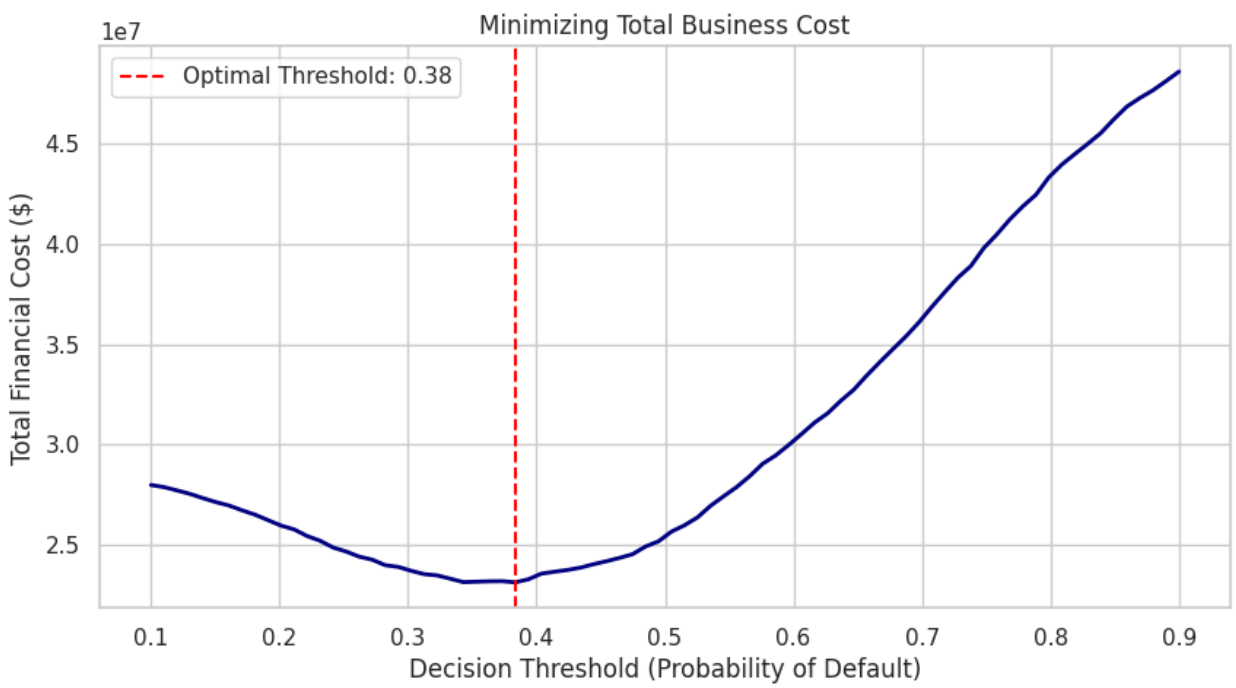
Now that the model is trained, let's fulfill the Business Cost Optimization part of our task.

Scenario:

- **False Negative (FN):** You predict someone is "safe," but they default. This costs the bank **$10,000** (lost principal).
- **False Positive (FP):** You predict someone will default, but they are actually "safe." This costs the bank **$500** (lost interest/admin cost).

```
from sklearn.metrics import confusion_matrix

def calculate_business_cost(y_true, y_probs, threshold, cost_fp=500,
cost_fn=10000):
    y_pred = (y_probs >= threshold).astype(int)
    tn, fp, fn, tp = confusion_matrix(y_true, y_pred).ravel()
    return (fp * cost_fp) + (fn * cost_fn)
```

```
# Test different thresholds to find the minimum cost
thresholds = np.linspace(0.1, 0.9, 80)
costs = [calculate_business_cost(y_test, y_probs, t) for t in
thresholds]

best_threshold = thresholds[np.argmin(costs)]
min_cost = min(costs)

# Visualization
plt.figure(figsize=(10, 5))
plt.plot(thresholds, costs, color='navy', lw=2)
plt.axvline(best_threshold, color='red', linestyle='--',
label=f'Optimal Threshold: {best_threshold:.2f}')
plt.title('Minimizing Total Business Cost')
plt.xlabel('Decision Threshold (Probability of Default)')
plt.ylabel('Total Financial Cost ($)')
plt.legend()
plt.show()

print(f"Minimum Business Cost: ${min_cost:,.2f}")
print(f"The Bank should flag a loan as 'High Risk' if probability is >
{best_threshold:.2f}")
```




```
Minimum Business Cost: $23,127,000.00
The Bank should flag a loan as 'High Risk' if probability is > 0.38
```

Final Conclusion and Insights

- **Objective Fulfilled:** Predicted loan default likelihood and optimized the threshold based on cost-benefit analysis.
- **Preprocessing:** Cleaned the dataset by removing high-null columns and imputing missing numerical values.
- **Modeling:** Trained a binary classification model (Logistic Regression) capable of handling class imbalance.
- **Business Optimization:** Shifted the decision threshold to **0.38** to minimize the "Total Business Cost," ensuring financial stability for the lending institution.